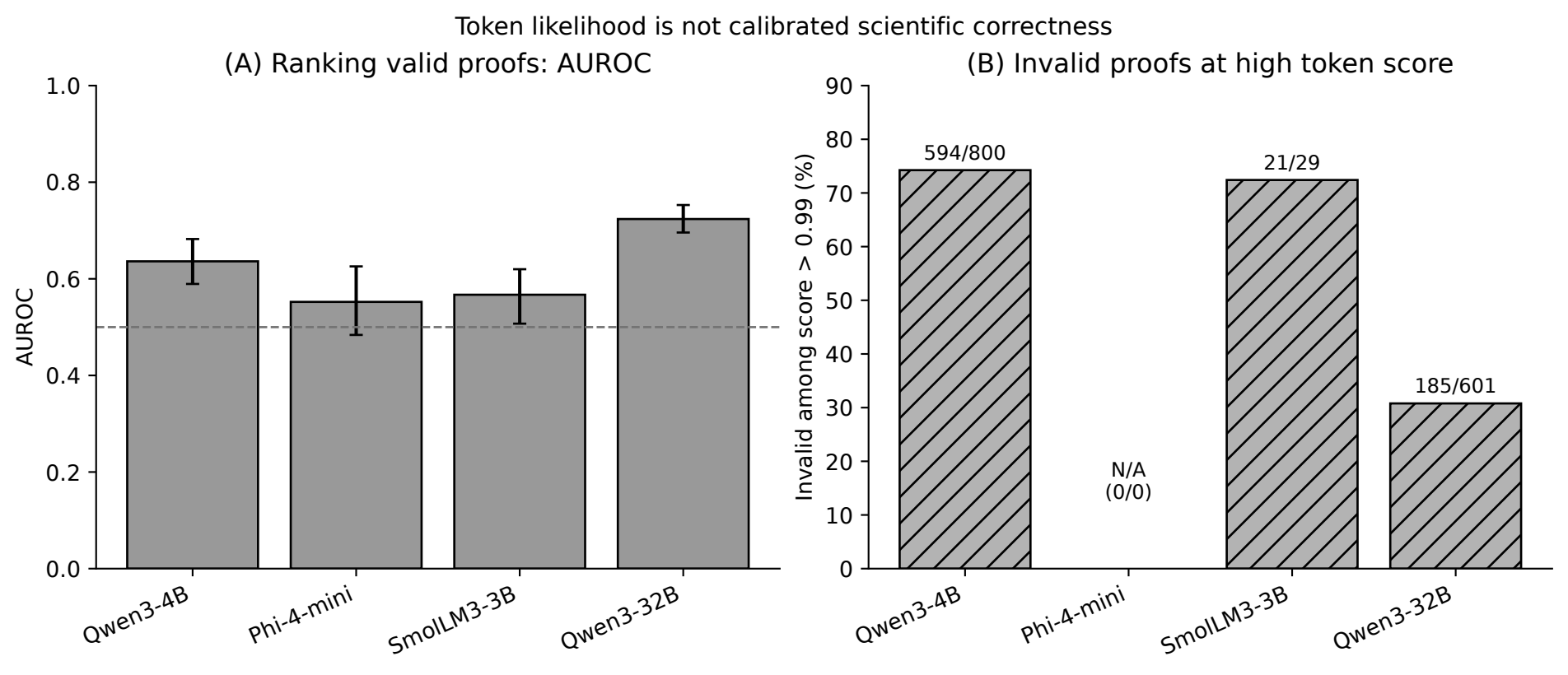
Token likelihood is not calibrated scientific correctness
(A) Ranking valid proofs: AUROC
AUROC
1.0
0.8
0.6
0.4
0.2
0.0
Qwen3-4B
Phi-4-mini
SmolLM3-3B
Qwen3-32B
(B) Invalid proofs at high token score
Invalid among score > 0.99 (%)
90
80
70
60
50
40
30
20
10
0
594/800
N/A
(0/0)
21/29
185/601
Qwen3-4B
Phi-4-mini
SmolLM3-3B
Qwen3-32B